

# Midterm # I

## CBE450: Process Design - Spring 2026

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### Problem 1: Manure Management (120 pts)



A dairy farm located in Eau Claire produces 1000 tons of waste manure and wants to get rid of it. The farmer is ambivalent about the value of its manure (it is willing to give it for free).

There is a corn farmer located in Menomonie that is willing to pay 0.5\$/ton for the manure and there is another soy bean farmer located in Black River Falls that is willing to pay 1.5\$/ton. These farmers want the manure because it is a valuable fertilizer that contains water, organic matter, and nutrients (phosphorus and nitrogen). The transportation cost for hauling manure to Menomonie is 0.1\$/ton and to Black River Falls is 0.2\$/ton. Each of these farmers can take at most 500 tons of manure.

All the farmers are part of a cooperative organization that works together to determine the best strategy to exchange products between them by solving a supply chain optimization problem.

Address the following:

1. (30 pts) Write an optimization problem in Julia to determine the optimal allocation of products for all stakeholders that maximizes the total profit. Draw a diagram of the supply chain to show the stakeholders, links and flows, optimal allocations; verify that all balances hold. Determine the total profit, supply/transport costs, and demand revenues of the system. Explain the nature of the optimal solution based on simple economic intuition (e.g., check the economic viability of the different pathways). Can the dairy farmer get rid of all its manure? Why or why not?
2. (30 pts) The Wisconsin Department of Natural Resources (DNR) has become increasingly concerned about the high concentration of nutrients in the soil near the Menomonie farm. This excess is the result of overapplying manure as fertilizer and triggers algae blooms in the local lake. In order to protect the lake, the DNR is introducing a new policy that asks the Menomonie farmer to pay 1.0\$/ton for any manure applied as fertilizer to cover lake remediation costs. As such, the Menomonie farmer is now forced to request manure at  $-0.5\$/ton$  (it is willing to take manure, but it charges for this service). This setting is similar to how a landfill works (a landfill is willing to take waste, but charges a tipping fee). Determine the optimal allocations for the stakeholders under this new setting and show them in a diagram. Determine the total profit, supply/transport costs, and demand revenues of the system. Explain the nature of the optimal solution based on simple intuition. Why can't the dairy farmer get rid of all its manure under this new setting? Was the DNR able to divert manure away from the Menomonie farm using the new policy?
3. (30 pts) Because the dairy farmer can no longer get rid of its manure under the new policy of the DNR, the dairy farmer is now *willing to pay* anyone that takes its manure 0.7\$/ton. Determine the optimal allocations and flows under this new setting and show them in a diagram. Determine the total profit, supply/transport costs, and demand revenues of the system. Explain the nature of the solution

based on simple intuition. Can the dairy farmer get rid of all its manure in this new setting? Why or Why not?

4. (30 pts) A new consumer of compost has appeared in Madison and is joining the cooperative; this consumer is quite far from Eau Claire (transportation cost of 1 \$/ton) but is willing to pay \$100 per ton of compost (compost is quite valuable in urban areas because it can be used in home gardens). This consumer can take up to 100 tons of compost. The dairy farmer in Eau Claire has a technology that can produce compost from its manure. This technology can generate 0.1 ton of compost per ton of manure, can process up to 500 tons of manure, and the operating cost is \$1 per ton of manure processed. Determine the optimal allocations, flows, and show them in a diagram. Determine the total profit, costs, and revenues of the system. Explain the nature of the solution based on simple intuition. Can the dairy farmer get rid of all its manure in this new setting? Compare the total profit of this setting with that of Question 3; did using the new technology increase the total profit of the system? If so, by how much?

**Note:** To solve this problem, it is not necessary to use any binary variables and it is not necessary to use any shadow prices (dual variables).